Ryan Rulkens

linkedin.com/in/ryan-rulkens-36106b237/ ▪ (925) 663-9633 ▪ ryanrulkens@gmail.com

OBJECTIVE

Motivated and energetic undergraduate student seeking an internship that integrates data analysis and ML tools to solve real-world problems. Experienced in Python, Java, and frameworks such as PyTorch and TensorFlow, with projects spanning neural network modeling, generative AI, large-scale data processing, and algorithm design. Hardworking and reliable team member. Available to work June 15th to September 18th.

EDUCATION

University Of California, San Diego — San Diego, CA

B.S. Data Science, B.S. Bioinformatics, GPA: 3.95 — **Expected June 2028**

- Relevant Coursework: Data Structures and Algorithms, Statistical Methods, Theoretical Foundations of Data Science, Practice and Application of Data Science, Practice and Application of Data Science Genetics, Organismic and Evolutionary Biology, Linear Algebra, Calculus and Analytic Geometry
- Notable project: Developed the API for a pandas-style dataframe in Java, using a hashtable for label-data mapping and linked lists for efficient storage and retrieval

TECHNICAL SKILLS

- ***Languages*: ** Python, Java, MATLAB, SQL, R
- ***Frameworks*: ** Pandas, Numpy, Pytorch
- ***Tools*: ** Tableau, Git, Jupyter, Excel, PowerBI

RELEVANT EXPERIENCE

Research Assistant, Alzheimer's Prevention Lab at UC San Diego — **Oct. 2025 – Present**

- Use python and R to clean, merge, and perform EDA on genomic, clinical, and biomarker data from the HABS HD (Health and Aging Brain Study) dataset
- Aid in manuscript preparation for lab publications

Teaching Assistant, BILD 1: The Cell — **January 2026 – Present**

- Lead discussions, hold office hours, create course materials, proctor midterms and exams

Music Recommendation App Development — **June – Sept. 2025**

- Processed multiple song datasets into a unified database in duckDB using Python
- Embedded lyrics for 100000+ songs with a miniLM sentence-transformer model, comparing cosine similarity with text embeddings client resumes to create customized song recommendations
- Developed a web app allowing clients to obtain best match recommendations

Machine Learning in Exoplanet Detection — **Summer 2021**

- Trained CNN and MLP models on star flux data collected from Kepler space telescope, utilizing Keras, TensorFlow, Pandas, NumPy, and sci-kit learn libraries
- Predicted existence of exoplanets in other star systems with testing accuracy of 99.7%

Genomic Analysis for Bioremediation Pathways — **May 2025**

- Applied bioinformatics tools (prodigal, eggnog mapper etc) as well as data processing and modelling in python to analyze genomic datasets on oceanic prokaryotes from Tara Oceans dataset
- Identified proteins with bioremediative potential, while gaining insight into taxonomy of organisms

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Workshops Board Member, UCSD Data Science Student Society — **Sept. 2024 – Present**

- Attended and hosted workshops in machine learning, AI, and data science applications in healthcare and finance

Beginner Programming Competition Committee Member, UCSD Women in Computing — **Sept. 2024 – Present**

- Wrote problems, created context, and added test cases for Fall Beginner Coding Competition

Undergraduate Bioinformatics Club, UCSD — **Sept. 2024 – Present**

- Wrote code to model biological processes (ex. action potentials)

AWARDS AND INTERESTS

- ***Honors and Awards*: ** National Merit Finalist (2023); 1st, Canadian International Music Competition (2023 and 2024); 6th, UCSD spring coding competition (2025); AIME Qualifier; Gold President's Volunteer Service Award
- ***Interests*: ** Music, Rock climbing and biking, Psychology, Strategy games, Investing, crochet